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HSC 250: Public Health Biology

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November 16, 2025

Chronic Obstructive Pulmonary Disease (COPD) is a progressive, non-communicable disease that significantly threatens population health worldwide. COPD is characterized by persistent airflow limitation, chronic inflammation, and irreversible lung tissue damage. It most commonly encompasses two conditions: chronic bronchitis, defined by a long-term productive cough caused by airway inflammation, and emphysema, which involves the destruction of alveoli responsible for oxygen exchange. Over time, individuals with COPD experience shortness of breath, wheezing, reduced exercise tolerance, and overall diminished quality of life. According to the *Global Initiative for Chronic Obstructive Lung Disease (GOLD)*, tobacco smoke, air pollution, occupational exposure, and household biomass fuels are the primary contributors to COPD development (“Global Strategy”). These risk factors result in chronic inflammation that permanently damages lung structures, making COPD a critical topic for public health intervention.

The burden of COPD on society is profound. The World Health Organization identifies COPD as the third leading cause of death globally, accounting for more than 3.2 million deaths each year (“Chronic Obstructive Pulmonary Disease”). Despite global health efforts, data show that the overall burden of COPD has remained high and, in some regions, continues to rise. The *Global Burden of Disease Study* reveals that COPD contributes more than seventy-four million disability-adjusted life years (DALYs) worldwide, reflecting both premature mortality and years lived with disability (Wang et al.). In the United States alone, the economic impact is substantial. The American Lung Association reports that COPD costs exceed forty-nine billion dollars

annually, accounting for direct medical care, hospital admissions, and loss of productivity (“COPD Trends Brief”). These economic and population-level burdens demonstrate the enormous strain COPD places on healthcare systems, families, and communities. Furthermore, COPD disproportionately affects older adults, individuals exposed to secondhand smoke, and people living in areas with poor air quality. Research also shows significant socioeconomic disparities: people with limited access to healthcare often receive delayed diagnoses or inadequate treatment, worsening long-term outcomes (Vianello and Guazzi). Altogether, the epidemiological, economic, and equity dimensions of COPD underscore why it remains a major non-communicable disease requiring sustained global and national response.

The peer-reviewed case study examined for this assignment provides an in-depth look at how COPD affects individuals at the personal and behavioral level. Gold and Ross conducted a qualitative case study titled “A Case Study of Chronic Obstructive Pulmonary Disease Self-Management in an Older Adult,” focusing on a seventy-two-year-old male patient recently discharged from the hospital following an acute COPD exacerbation. The authors aimed to understand how the patient managed his illness at home, what challenges he faced in following clinical recommendations, and how environmental, emotional, and behavioral factors influenced his overall health. The study design incorporated structured interviews, direct observation, and review of medical records, allowing the researchers to capture both quantitative data (such as medication lists and hospital history) and qualitative insights regarding the patient’s thoughts, emotions, and habits (Gold and Ross).

The authors found that the patient struggled significantly with several aspects of self-management. First, despite believing he used his inhaler correctly, observation showed incorrect inhaler technique, particularly with timing and breath coordination. This error reduced the effectiveness of his prescribed medications. Second, although he successfully stopped smoking after hospitalization, he expressed extreme anxiety about relapse, demonstrating the psychological burden associated with long-term smoking cessation. Third, his home environment contributed to symptom worsening: the apartment contained dust, limited airflow, and restricted space for physical activity. These environmental factors made it difficult for him to perform pulmonary rehabilitation exercises, which are essential for maintaining lung function. Finally, the patient lacked structured self-monitoring strategies. He recorded symptoms occasionally, but he did not use a validated COPD action plan or regularly track peak flow measurements. The study highlights how environmental, emotional, and behavioral barriers — beyond the disease itself — can severely impact COPD outcomes.

My assessment of this case study is that it provides a clear, focused, and practical look at the lived experience of an older adult managing COPD. The authors effectively demonstrated that COPD is not only a biomedical diagnosis but also a condition shaped by environmental constraints, mental health, lifestyle factors, and health literacy. However, while the study is impactful, it has several limitations. First, the single-patient design limits generalizability. COPD patients vary widely in socioeconomic status, cultural background, and access to care; therefore, a larger sample or comparative cases could have provided stronger conclusions. Second, the

study did not include long-term follow-up, leaving uncertainty about whether the patient's difficulties improved, remained the same, or worsened over time. COPD is a chronic, progressive illness, so understanding long-term management would have strengthened the study. Third, the study did not incorporate modern telehealth or digital self-management tools. Research indicates that digital health interventions — such as apps that support inhaler reminders or tele-monitoring — can enhance COPD management among older adults (Nittari et al.). Including such elements could have provided a broader perspective on potential interventions. Lastly, the study could have examined the role of community resources, such as pulmonary rehabilitation programs or home-based nursing visits, which can be essential for patients with limited mobility. Although the study captured the patient's daily challenges well, evaluating the larger public health and community context would have made the findings more comprehensive.

From a public health standpoint, this case study offers valuable lessons. COPD management requires more than medical treatment; it demands a holistic approach that addresses environmental, psychological, and behavioral influences on health. Public health professionals can apply these insights by improving patient education, especially regarding proper inhaler use. Many patients, like the one in the case study, believe they use inhalers correctly when they do not, leading to preventable exacerbations. Additionally, public health agencies could support community-based environmental assessments to ensure patients live in spaces conducive to lung health. Mental health support is equally important. Anxiety, depression, and fear of breathlessness are common in people with COPD and often interfere with self-management.

Integrating mental health counseling or support groups into COPD care could strengthen patient outcomes. Furthermore, public health programs can encourage the use of self-monitoring tools and digital health technologies to help patients track symptoms more consistently. Overall, public health professionals can use the findings from this case study to advance more patient-centered interventions that address the social, environmental, and emotional challenges of COPD.

In conclusion, COPD remains a major global health issue due to its persistent prevalence, high mortality rate, and extensive economic costs. The disease is influenced by modifiable risk factors, social determinants of health, and lifelong exposures that place certain populations at greater risk. The case study reviewed in this paper shows that COPD management is complex and influenced by multiple factors beyond medication alone. By integrating patient education, environmental improvements, psychological support, and community resources, public health professionals can develop more effective responses to COPD. This holistic approach is critical for reducing the overall burden of this non-communicable disease and improving the quality of life for individuals affected by it.

References

- Chronic Obstructive Pulmonary Disease (COPD).” *World Health Organization*, 2023,
[https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease-\(copd\)](https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease-(copd))
- COPD Trends Brief: Burden.” *American Lung Association*, 2024,
<https://www.lung.org/research/trends-in-lung-disease/copd-trends-brief/copd-burden>
- Gold, M. J., and M. M. Ross. “A Case Study of Chronic Obstructive Pulmonary Disease Self-Management in an Older Adult.” *Clinical Nursing Studies*, vol. 7, no. 2, 2019, pp. 25–32.
- Nittari, Giulio, et al. “Telemedicine and COPD Self-Management: Improving Health Monitoring for Older Adults.” *Frontiers in Digital Health*, vol. 5, 2023,
<https://www.frontiersin.org/articles/10.3389/fdgth.2023.1144075/full>
- Vianello, Andrea, and Marco Guazzi. “Health Inequality in the Global Burden of COPD.” *COPD: Journal of Chronic Obstructive Pulmonary Disease*, 2021,
<https://www.dovepress.com/health-inequality-in-the-global-burden-of-chronic-obstructive-pulmonary-peer-reviewed-fulltext-article-COPD>
- Wang, Chen, et al. “Trends in COPD Burden and Risk Factors: Global Burden of Disease Study 2021.” *Respiratory Research*, 2024,
<https://respiratory-research.biomedcentral.com/articles/10.1186/s12931-024-03051-2>